Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2620586

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While some animal species exist outside of the usual male or female sexes, most species are substantially either male or female. Although many species exhibit a 1:1 sex ratio at birth, other species deviate from an even sex ratio. This is called adaptive sex ratio variation. For example, the temperature of the nest incubating eggs of the American alligator influences the sex ratios at birth.

The role of lampreys is complex. In some lake habitats, they are seen as parasites with a significant impact on the ecosystem, whereas lampreys are also a food source in some regions of the world, such as Scandinavia, the Baltics, and for some Indigenous peoples of the Pacific Northwest in North America.

The sex ratio of sea lampreys can vary based on external circumstances. Sea lampreys become male or female depending on how quickly they grow during the larval stage. These larval growth rates are influenced by the availability of food. In environments where food availability is low, growth rates will be lower, and the percentage of males can reach approximately 78population. In environments where food is more readily available, the percentage of males has been observed to be approximately 56

We focus on the question of sex ratios and their dependence on local conditions, specifically for sea lampreys. Sea lampreys live in lake or sea habitats and migrate up rivers to spawn. The task is to examine the advantages and disadvantages of the ability for a species to alter its sex ratio depending on resource availability. Your team should develop and examine a model to provide insights into the resulting interactions in an ecosystem.

Questions to examine include the following:

• What is the impact on the larger ecological system when the population of lampreys can alter its sex ratio?

• What are the advantages and disadvantages to the population of lampreys?

• What is the impact on the stability of the ecosystem given the changes in the sex ratios of lampreys?

• Can an ecosystem with variable sex ratios in the lamprey population offer advantages to others in the ecosystem, such as parasites?

Your PDF solution of no more than 25 total pages should include: • One-page Summary Sheet. • Table of Contents. • Your complete solution. • References list. • AI Use Report (If used does not count toward the 25-page limit.)

Note: There is no specific required minimum page length for a complete MCM submission. You may use up to 25 total pages for all your solution work and any additional information you want to include (for example: drawings, diagrams, calculations, tables). Partial solutions are accepted. We permit the careful use of AI such as ChatGPT, although it is not necessary to create a solution to this problem. If you choose to utilize a generative AI, you must follow the COMAP AI use policy. This will result in an additional AI use report that you must add to the end of your PDF solution file and does not count toward the 25 total page limit for your solution. Glossary Lampreys: Lampreys (sometimes inaccurately called lamprey eels) are an ancient lineage of jawless fish of the order Petromyzontiformes. The adult lamprey is characterized by a toothed, funnel-like sucking mouth. Lampreys live mostly in coastal and fresh waters and are found in most temperate regions.

Notations

Notation	Description
L	Total number of lampreys
M	Number of male lampreys
F	Number of female lampreys
r	Birth rate of lampreys
d	Mortality rate of lampreys
p	Probability of a female lamprey giving birth to a male

0.1 Mathematical Model

The dynamic of the Lamprey population can be described by the following differential equations:

$$\frac{dM}{dt} = rM(F) - dM$$
$$\frac{dF}{dt} = rF(M) - dF$$

where $rM(F)$ and $rF(M)$ are the birth rates of male and female lampreys, respectively. The mortality rates dM and dF account for natural deaths and other factors. The probability of a female lamprey giving birth to a male is represented by p.

1 Problem 1

Here we want to include the Lotka-Volterra Equation: